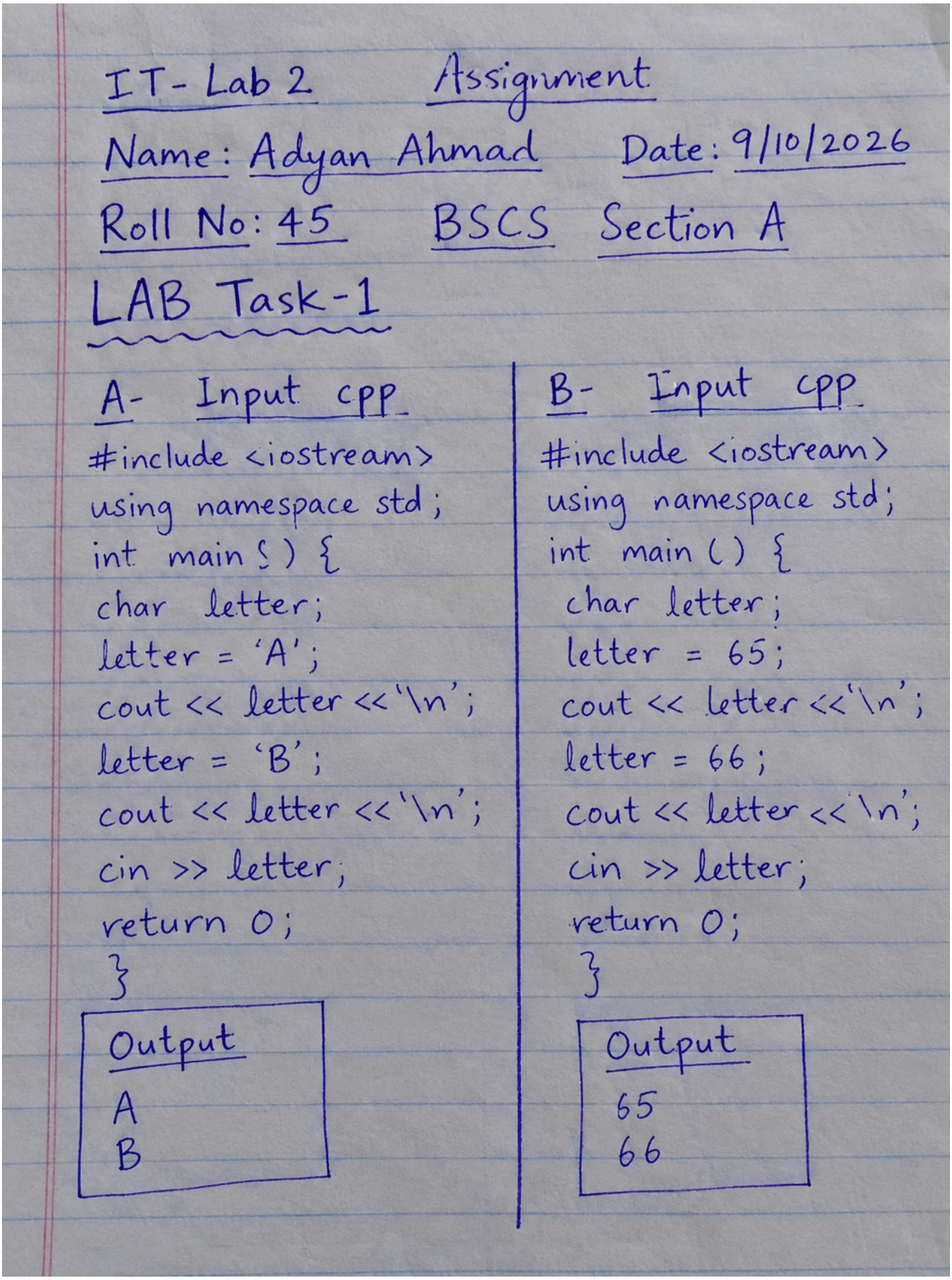
Lab Task 1

Handwritten code



IT- Lab 2 Assignment

Name: Adyan Ahmad Date: 9/10/2026

Roll No: 45 BSCS Section A

LAB Task-1

A- Input cpp

```
#include <iostream>
using namespace std;
int main() {
char letter;
letter = 'A';
cout << letter <<'\n';
letter = 'B';
cout << letter <<'\n';
cin >> letter;
return 0;
}
```

Output

A
B

B- Input cpp

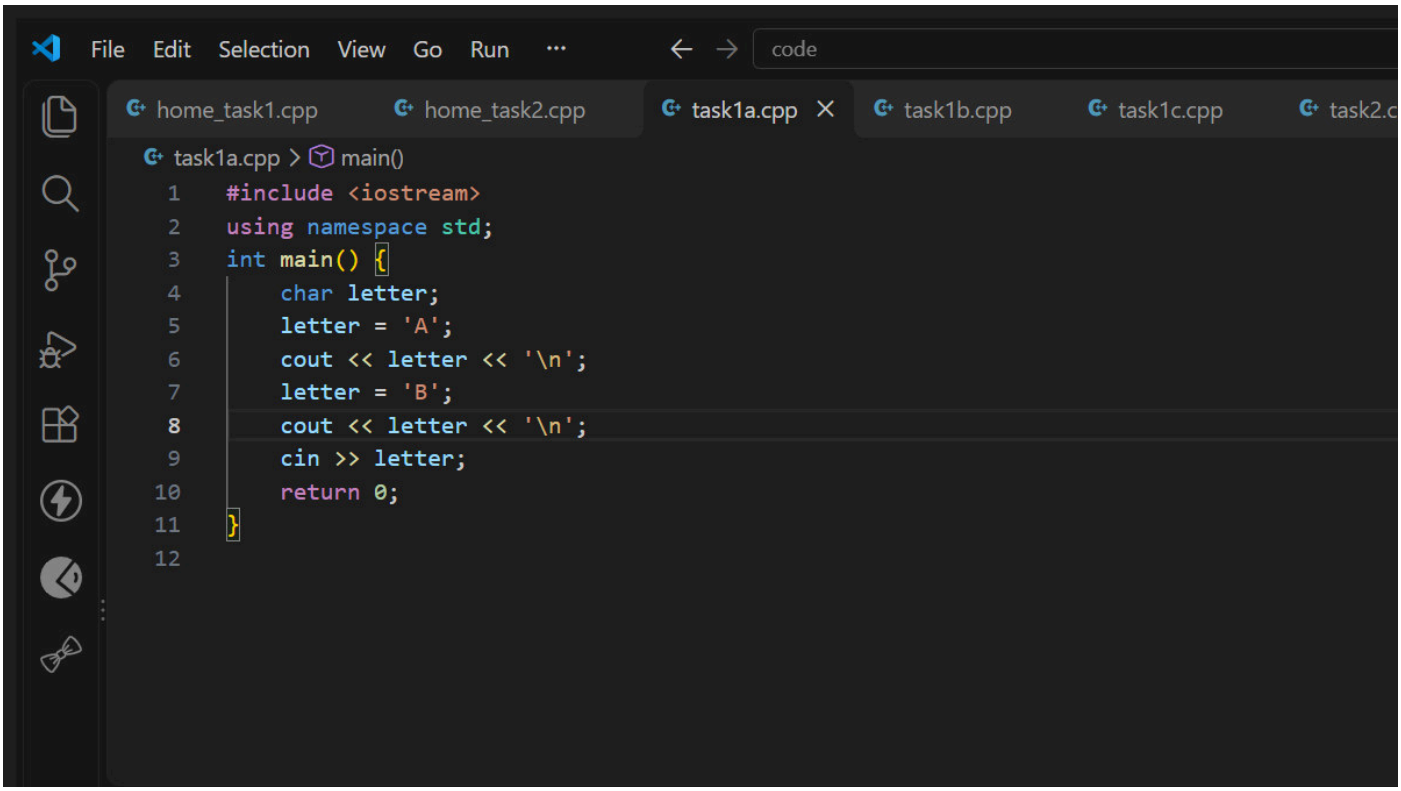
```
#include <iostream>
using namespace std;
int main () {
 char letter;
 letter = 65;
 cout << letter <<'\n';
 letter = 66;
 cout << letter <<'\n';
 cin >> letter;
 return 0;
 }
```

Output

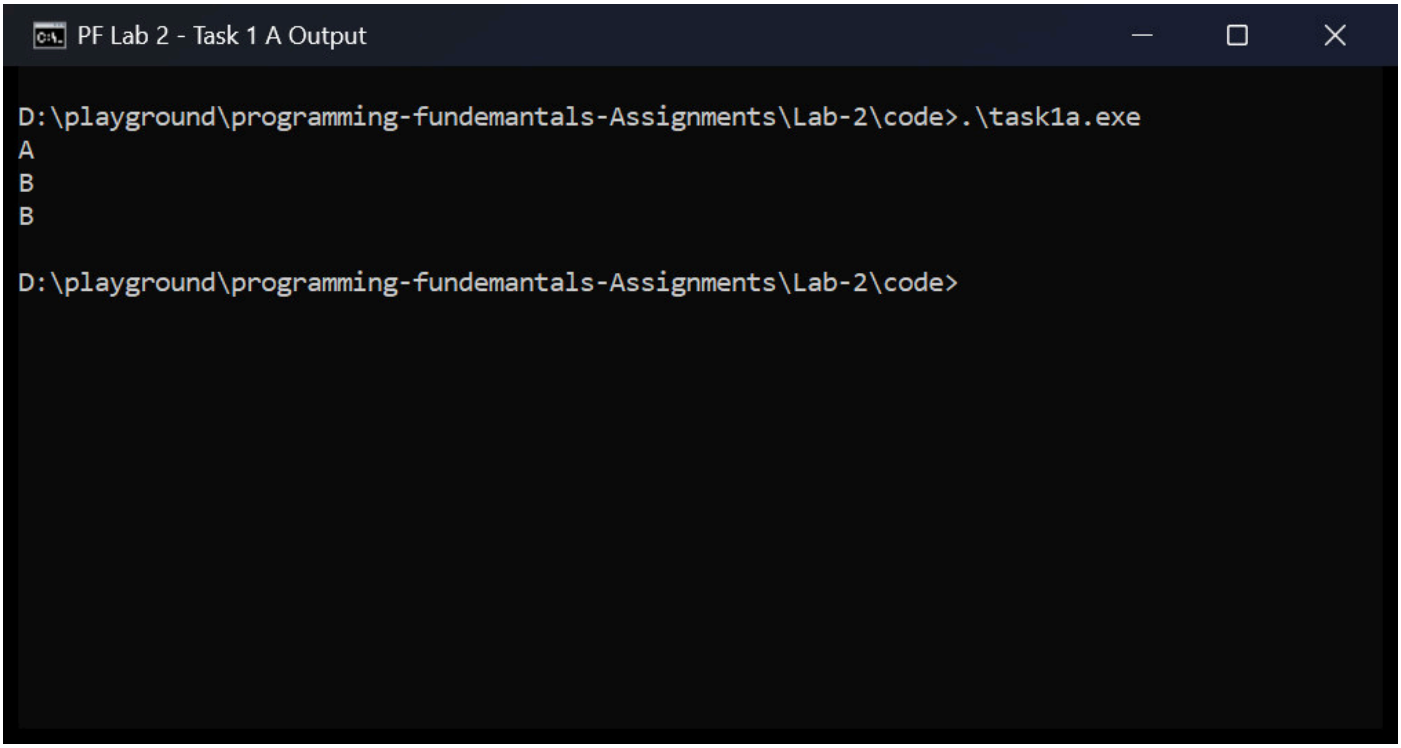
65
66

Lab Task 1 - Program A

Typed code (task1a.cpp)

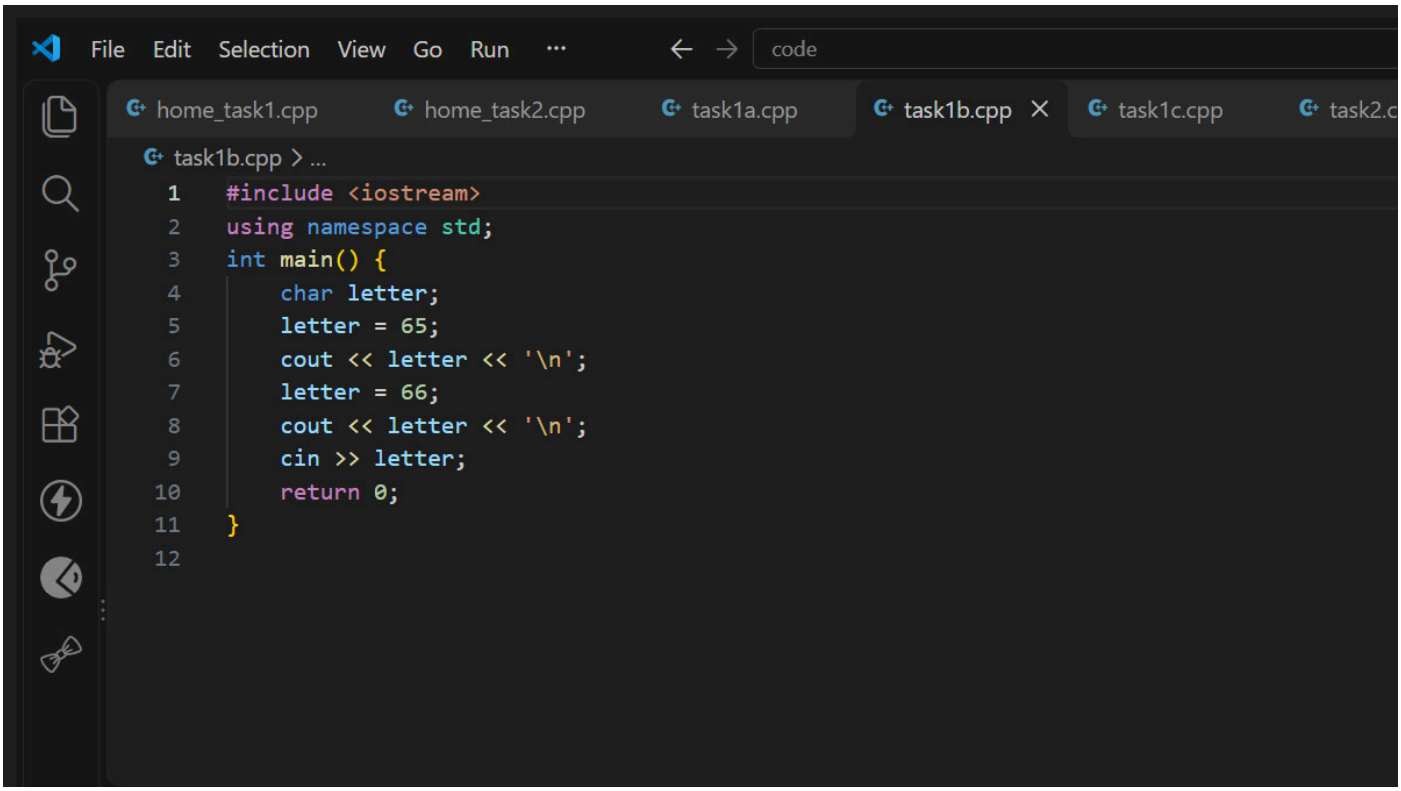


Output

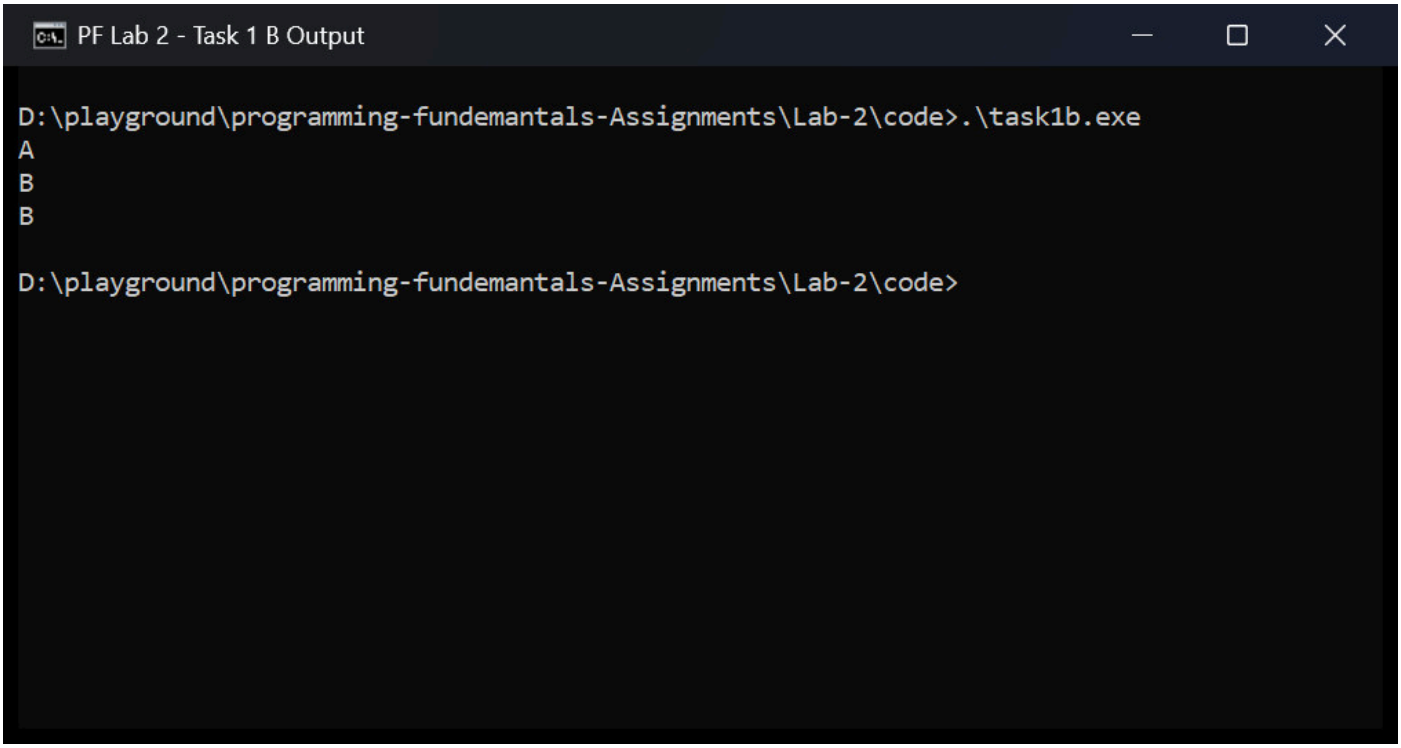


Lab Task 1 - Program B

Typed code (task1b.cpp)

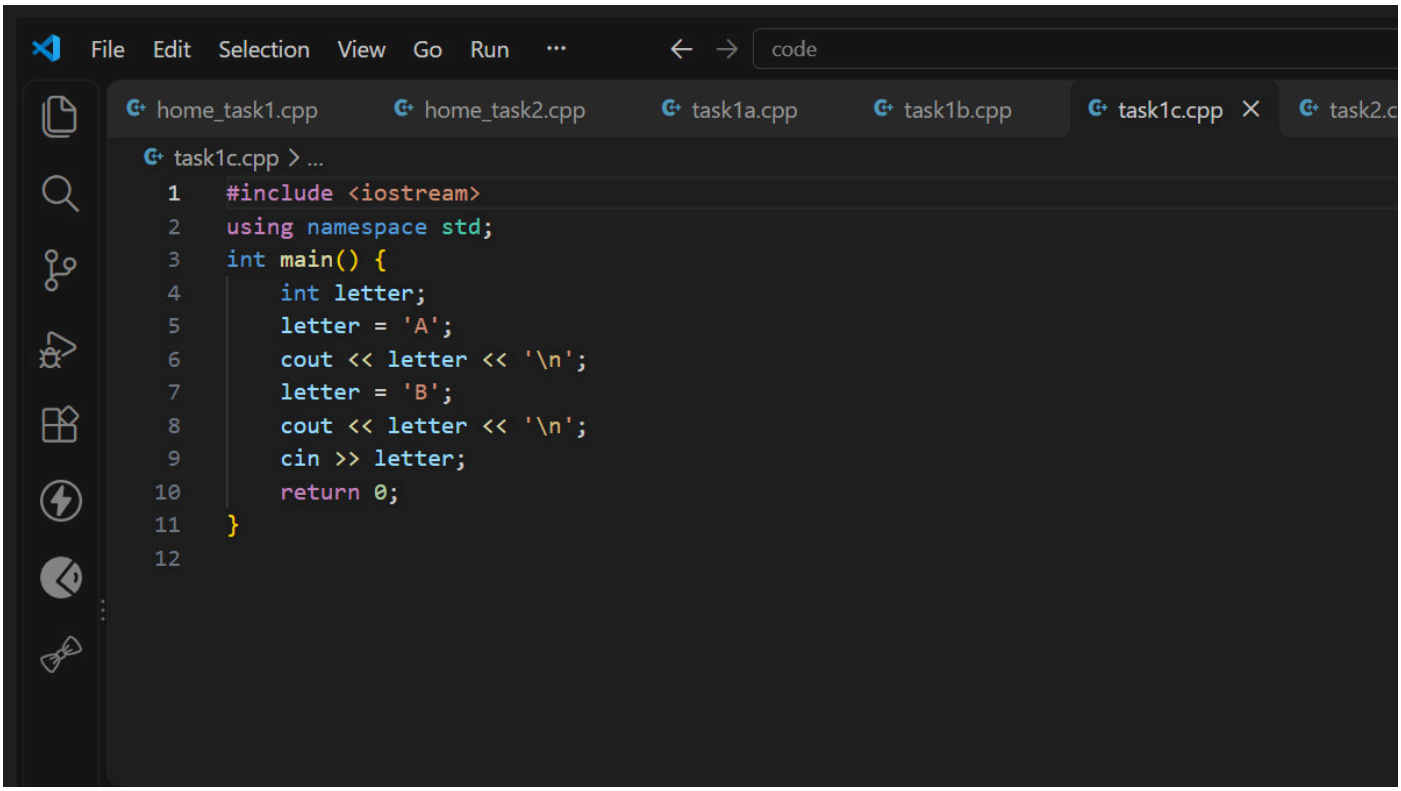


Output

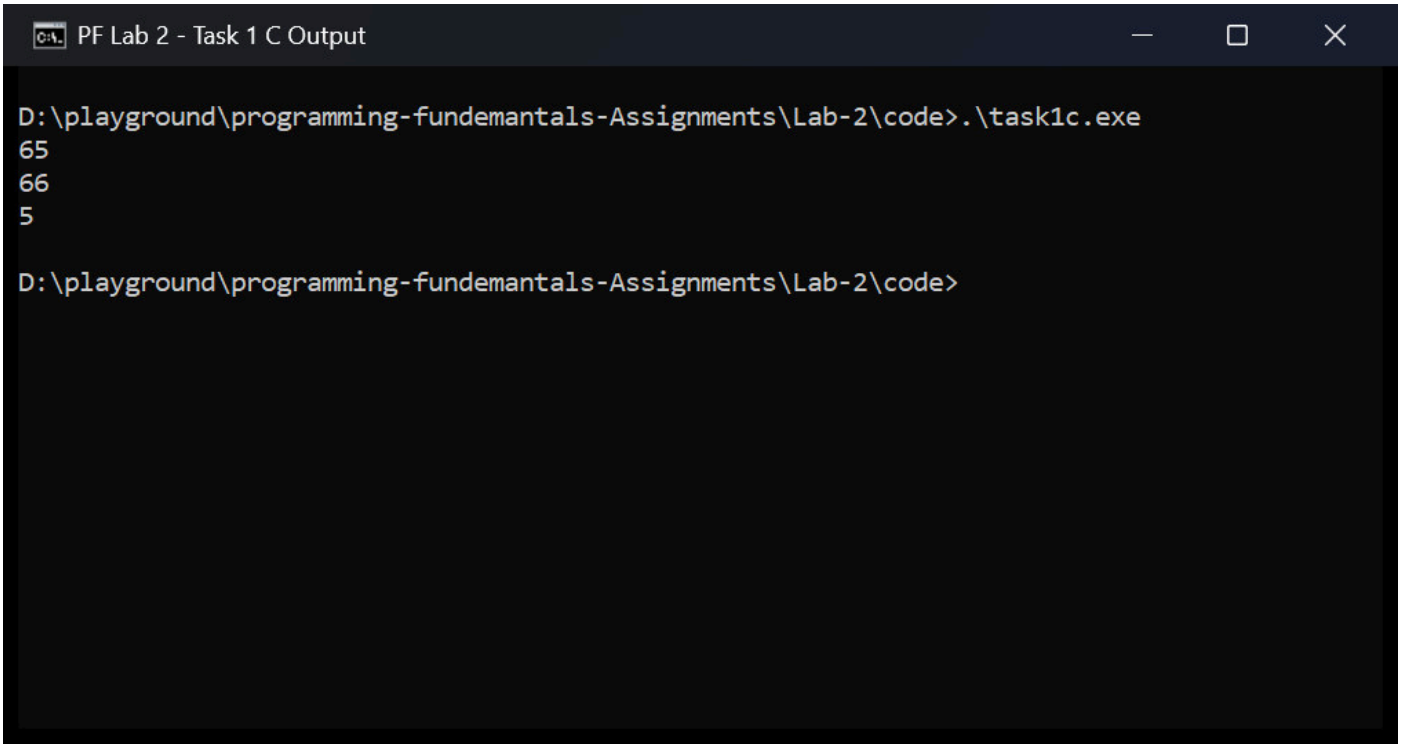


Lab Task 1 - Program C

Typed code (task1c.cpp)



Output



Lab Task 2

Handwritten code

Task 2:-

Source Code (Cpp)

```
#include <iostream>
using namespace std;
int main(){
  char letter;
  letter = 'A';
  cout << letter <<"\n";
  letter = 'B';
  cout << letter <<"\n";
  cin >> letter;
  return 0;
  }
```

Output

```
A
B
(after you type a letter)
the same letter will be
printed, for example:
B
```

Task 3:-

Source Code (Cpp)

```
#include <iostream>
using namespace std;
int main(){
  int val1;
  int val2;
  int val3;
  cout << "Please enter your 3 numbers:";
  cin >> val1 >> val2 >> val3;
  cout << val1 << "\n" << val2 << "\n" << val3 <<"\n";
  cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
  return 0;
  }
```

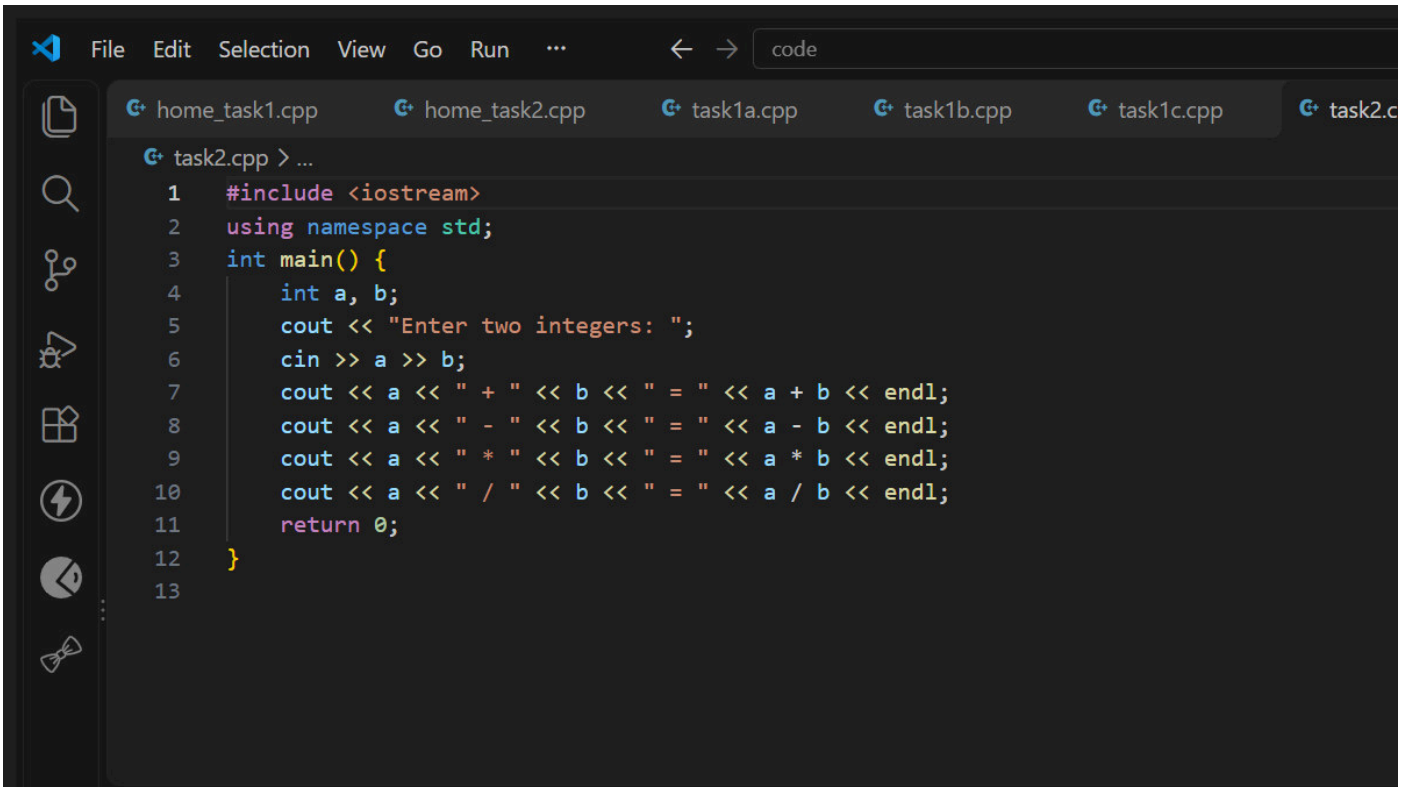
Output

```
Please enter your 3 numbers:
10 20 30
10
20
30

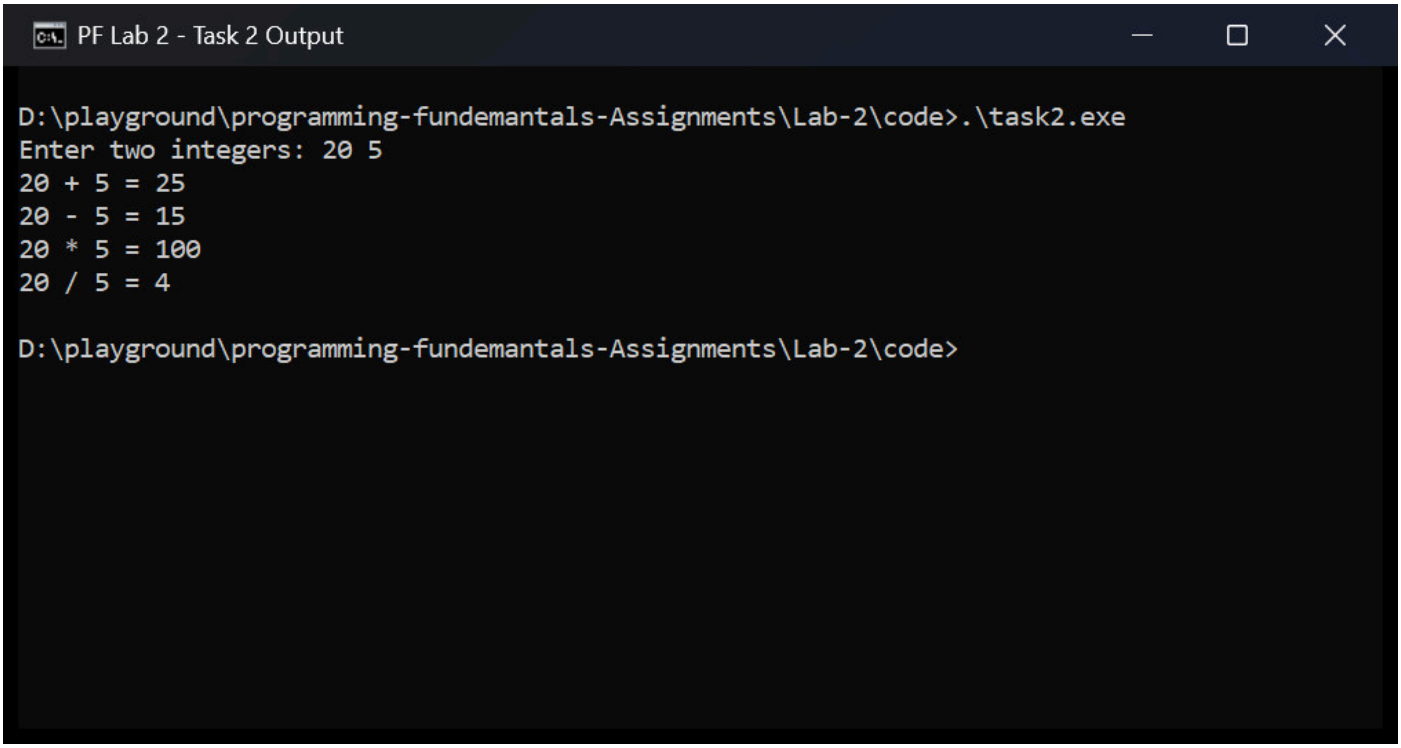
30
20
10
```

Lab Task 2

Typed code (task2.cpp)



Output



Lab Task 3

Handwritten code

Task 2:-

Source Code (Cpp)

```
#include <iostream>
using namespace std;
int main(){
  char letter;
  letter = 'A';
  cout << letter <<"\n";
  letter = 'B';
  cout << letter <<"\n";
  cin >> letter;
  return 0;
  }
```

Output

```
A
B
(after you type a letter)
the same letter will be
printed, for example:
B
```

Task 3:-

Source Code (Cpp)

```
#include <iostream>
using namespace std;
int main(){
  int val1;
  int val2;
  int val3;
  cout << "Please enter your 3 numbers:";
  cin >> val1 >> val2 >> val3;
  cout << val1 << "\n" << val2 << "\n" << val3 <<"\n";
  cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
  return 0;
  }
```

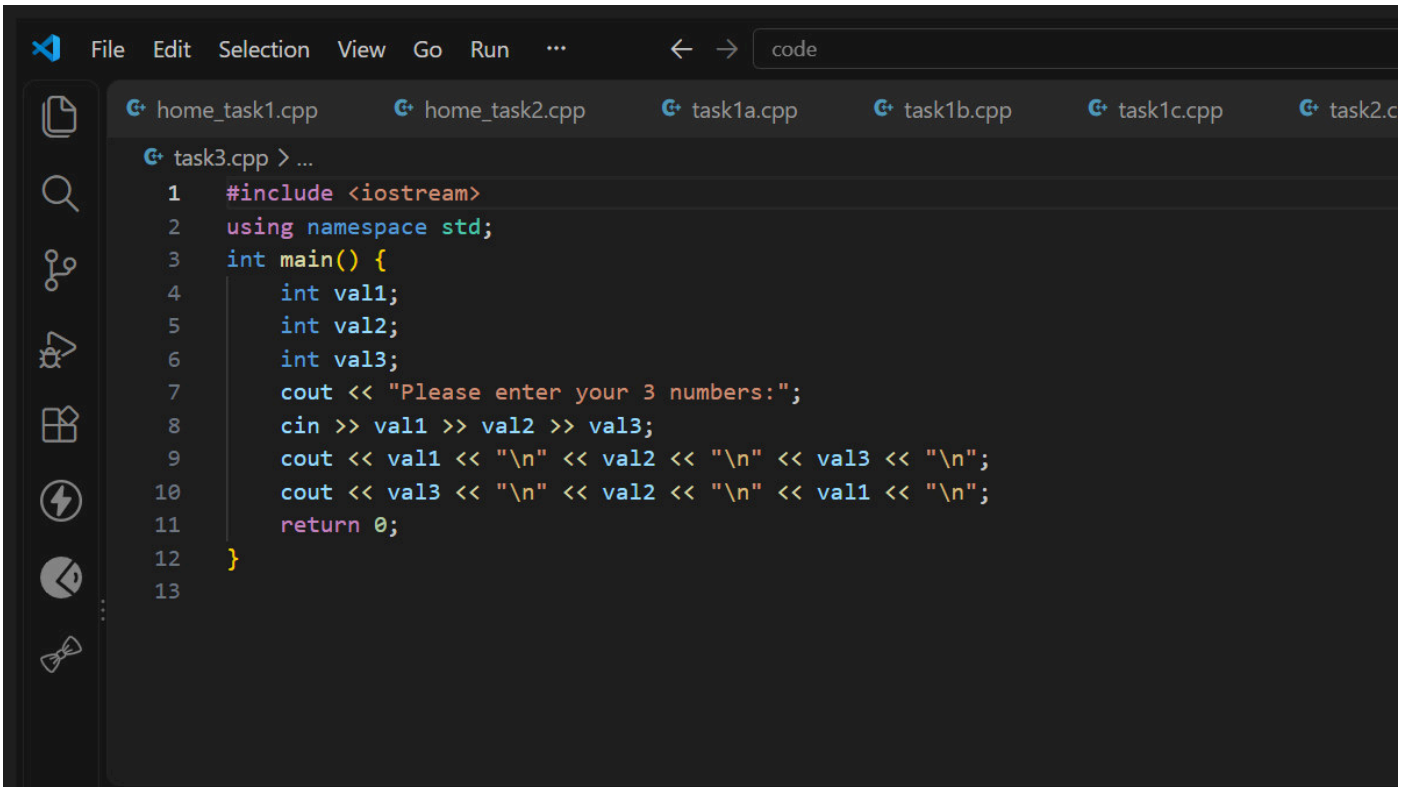
Output

```
Please enter your 3 numbers:
10 20 30
10
20
30

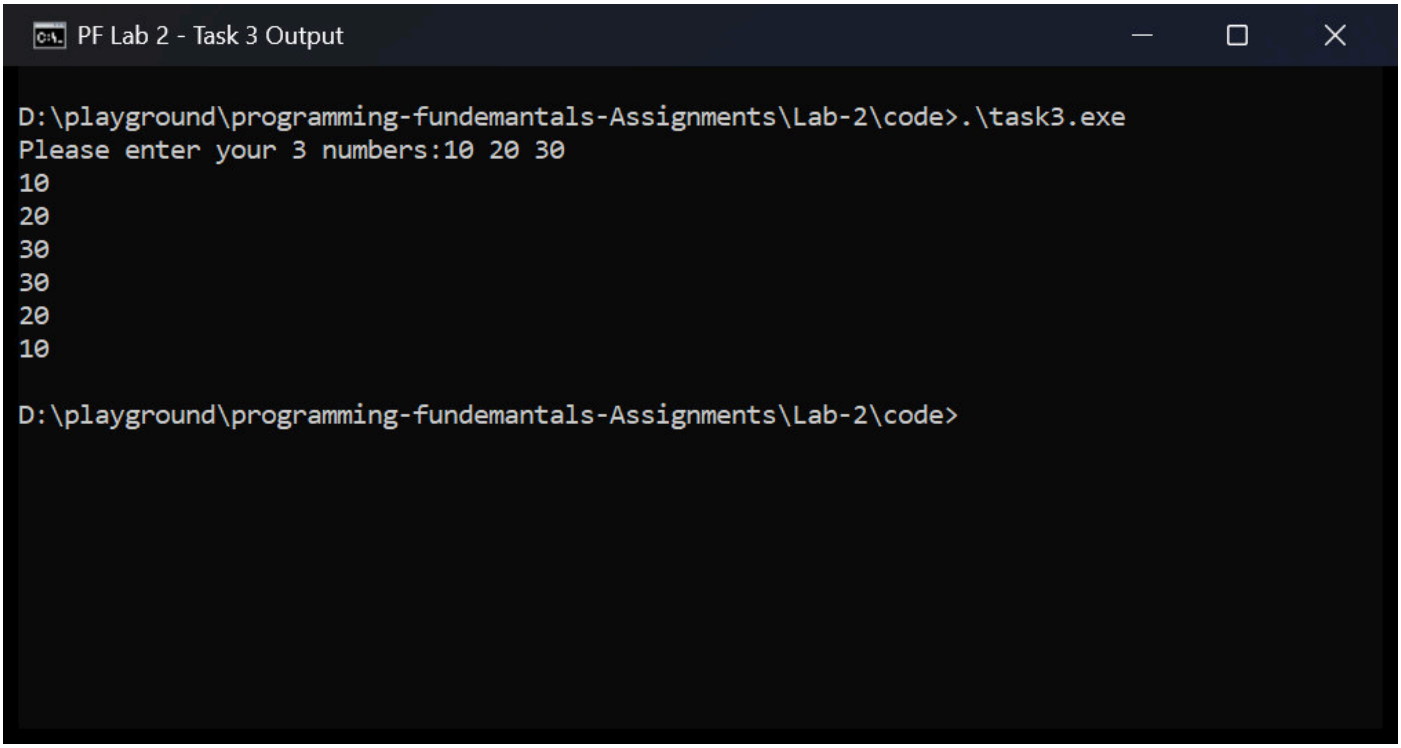
30
20
10
```

Lab Task 3

Typed code (task3.cpp)



Output



Lab Task 4

Handwritten code

Task 4:-

Source Code (cpp)

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main() {
  double x1, y1, x2, y2;
  cin >> x1 >> y1 >> x2 >> y2;
  double d = sqrt(pow(x2-x1, 2) + pow(y2-y1, 2));
  cout << d;
  return 0;
}
```

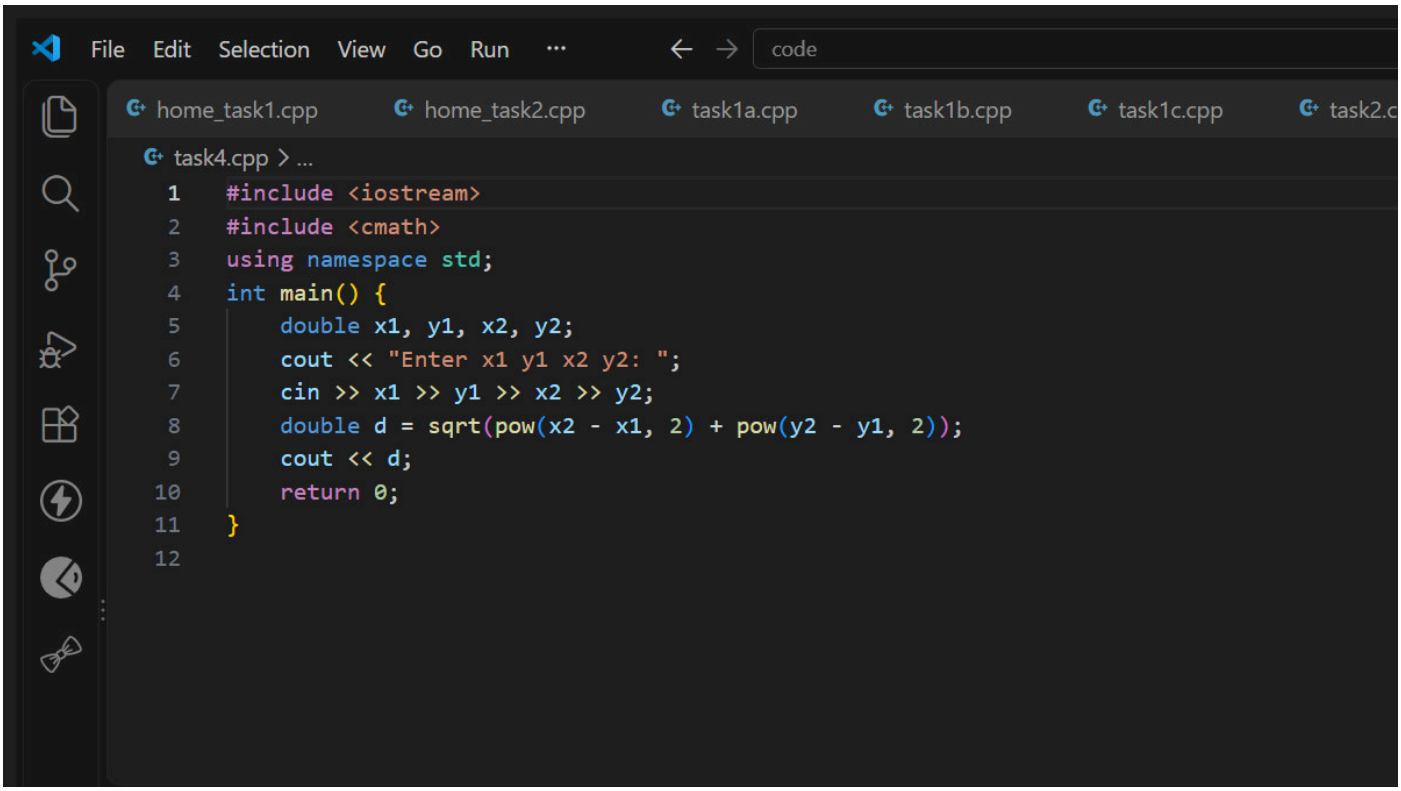
Output

```
Enter x1 y1 x2 y2:
0 0 3 4
5
```

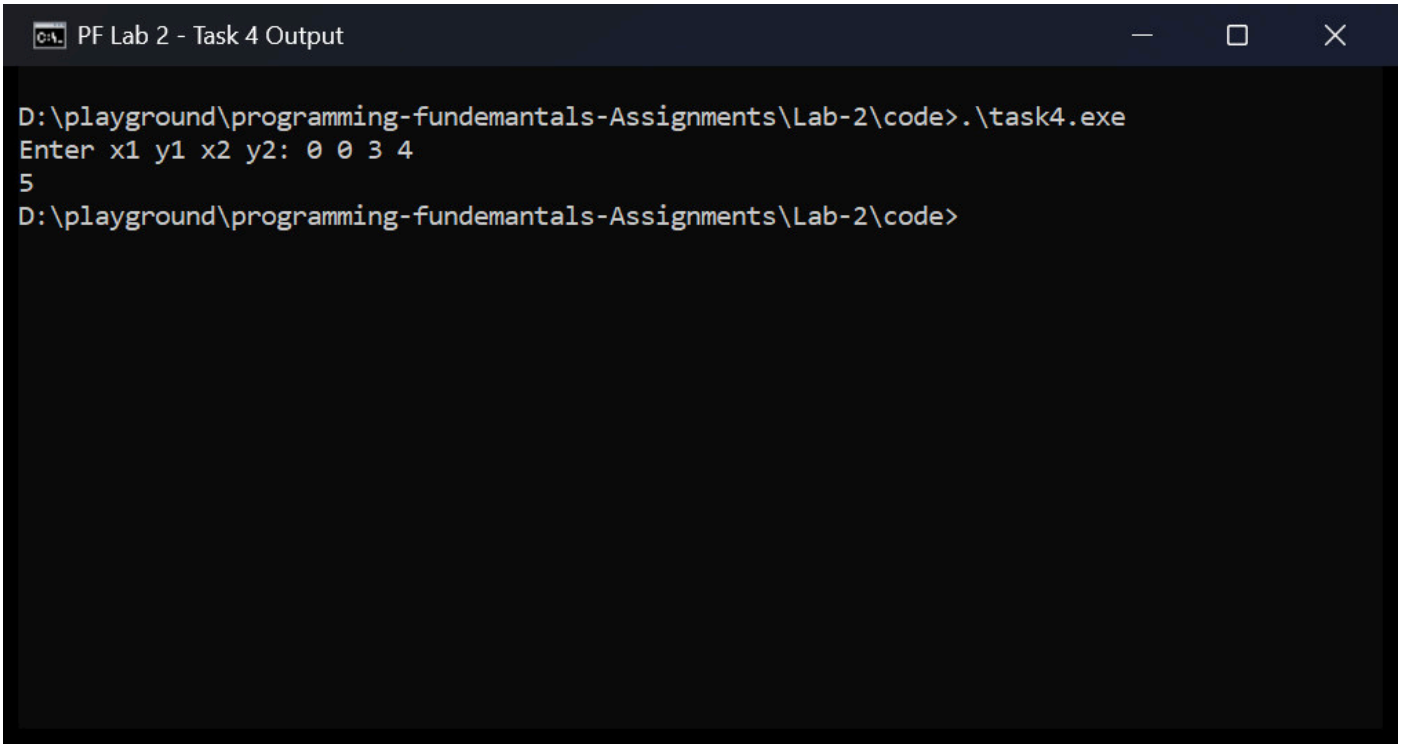
(Example input and output)

Lab Task 4

Typed code (task4.cpp)

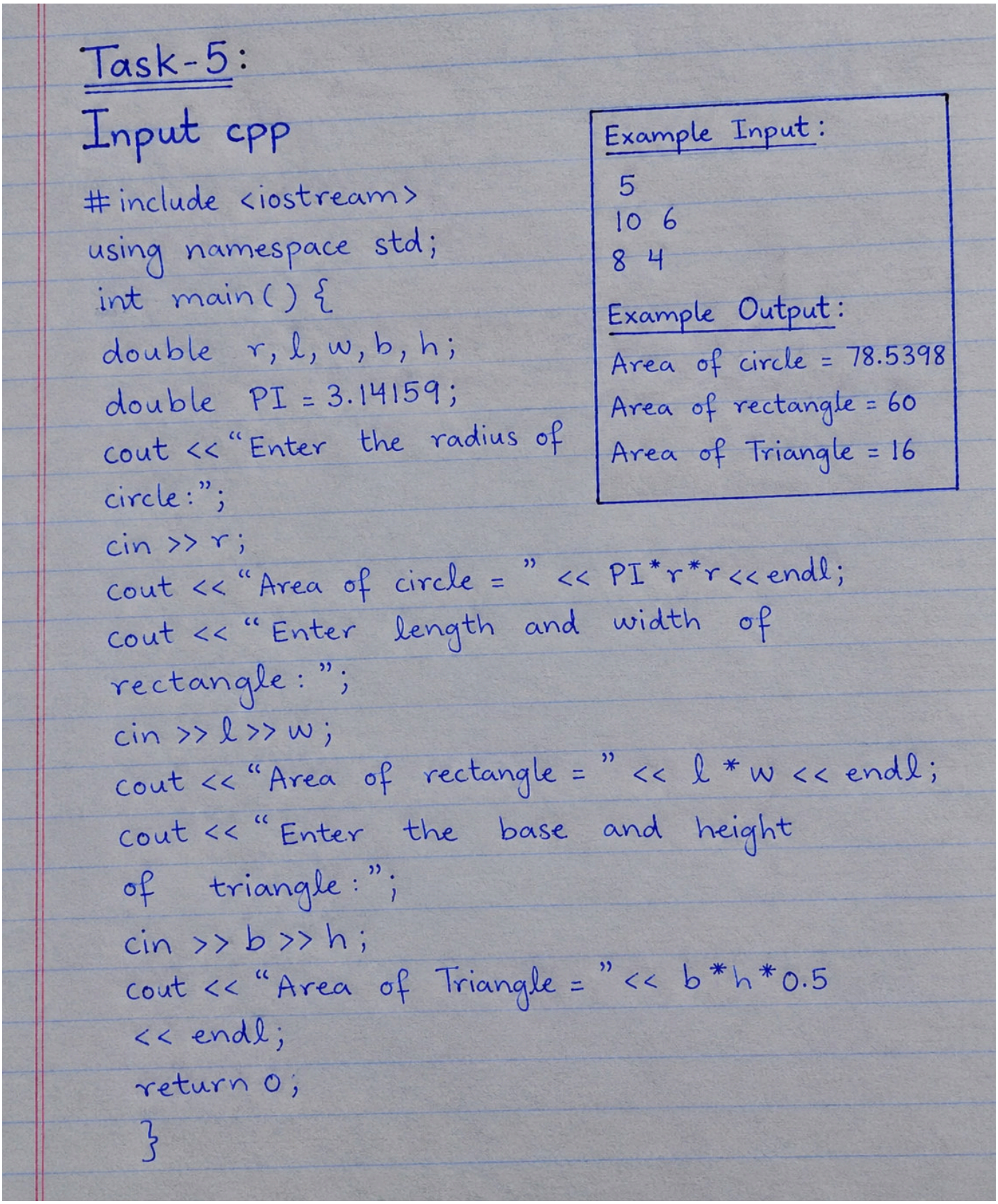


Output



Lab Task 5

Handwritten code



Task-5:

Input cpp

```
#include <iostream>
using namespace std;
int main() {
double r, l, w, b, h;
double PI = 3.14159;
cout << "Enter the radius of
circle:";
cin >> r;
cout << "Area of circle = " << PI*r*r << endl;
cout << "Enter length and width of
rectangle:";
cin >> l >> w;
cout << "Area of rectangle = " << l * w << endl;
cout << "Enter the base and height
of triangle:";
cin >> b >> h;
cout << "Area of Triangle = " << b*h*0.5
<< endl;
return 0;
}
```

Example Input:

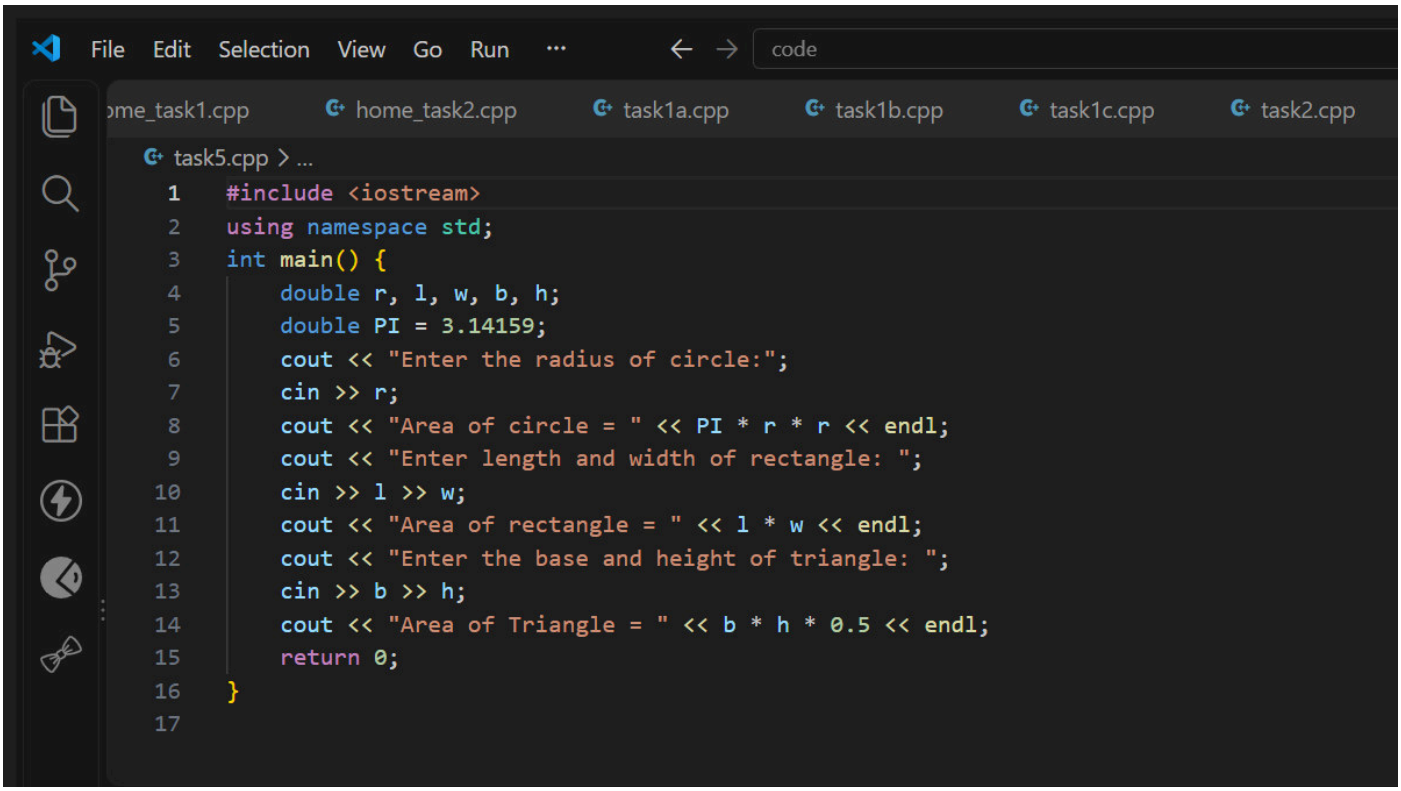
```
5
10 6
8 4
```

Example Output:

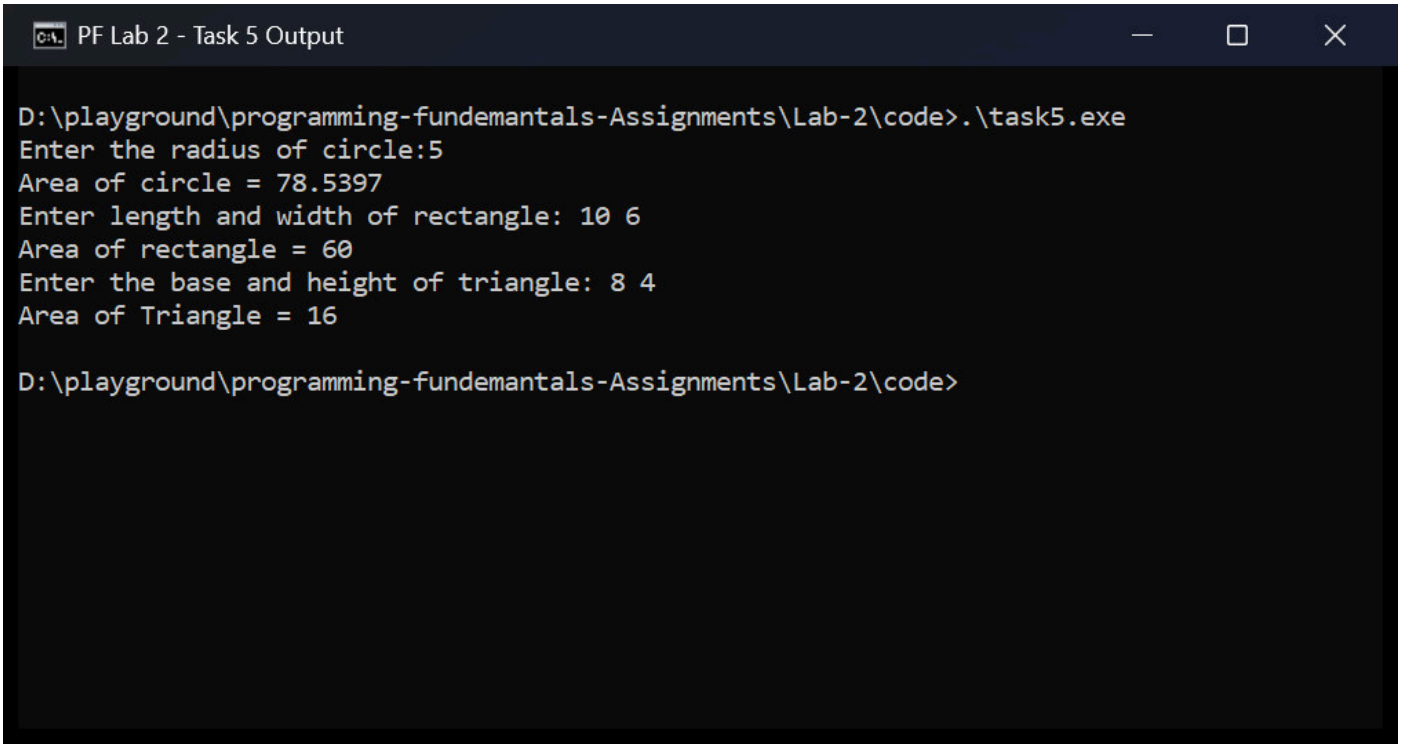
```
Area of circle = 78.5398
Area of rectangle = 60
Area of Triangle = 16
```

Lab Task 5

Typed code (task5.cpp)



Output



Lab Task 6

Handwritten code

Task-6:-

Source Code (cpp)

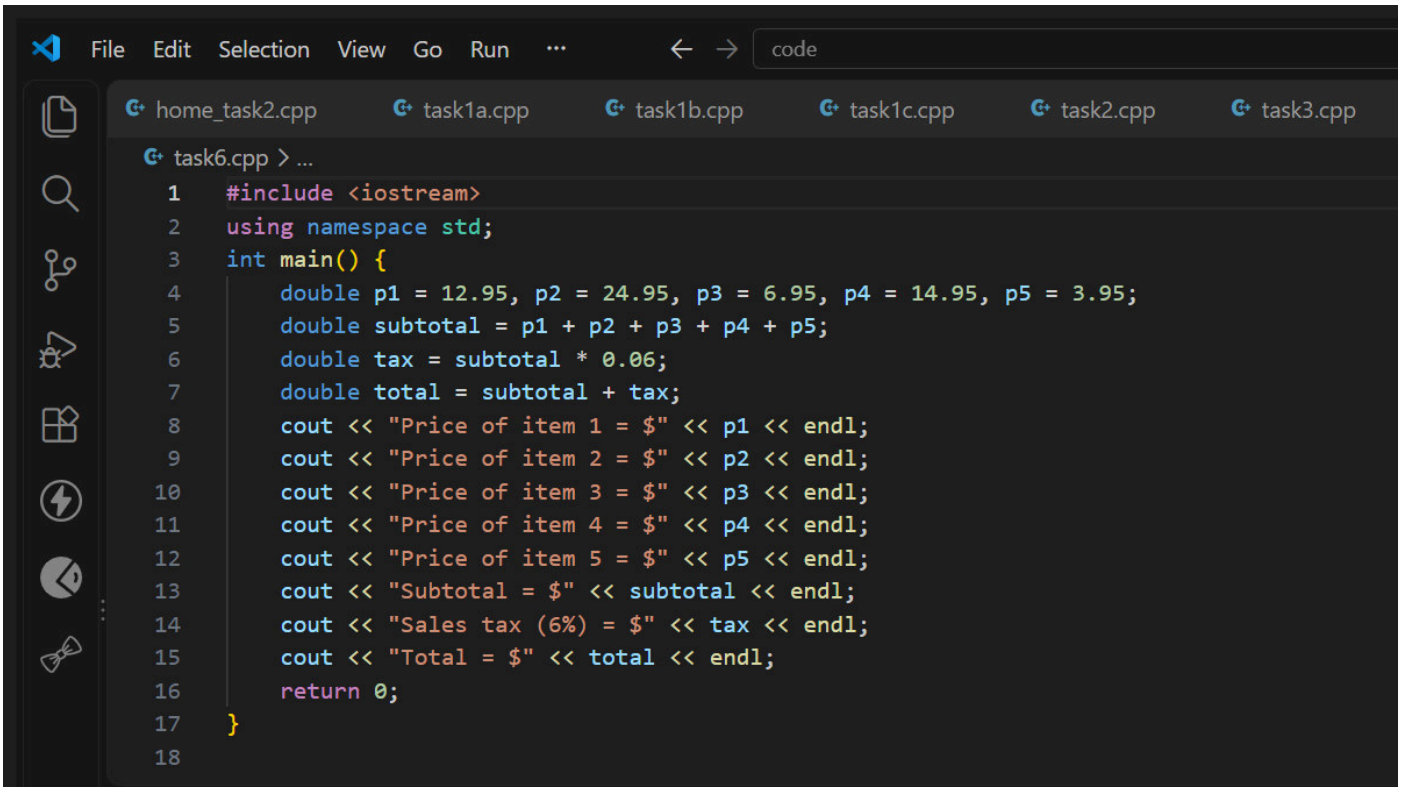
```
#include <iostream>
using namespace std;
int main () {
   double p1 = 12.95, p2 = 24.95, p3 = 6.95, p4 = 14.95, p5 = 3.95;
   double subtotal = p1 + p2 + p3 + p4 + p5;
   double tan = subtotal * 0.06;
   double total = subtotal + tan;
   cout << "price of item 1 = $" << p1 << endl;
   cout << "price of item 2 = $" << p2 << endl;
   cout << "price of item 3 = $" << p3 << endl;
   cout << "price of item 4 = $" << p4 << endl;
   cout << "price of item 5 = $" << p5 << endl;
   cout << "insubtotal = $" << subtotal << endl;
   cout << "Sales tan (6%) = $" << tan << endl;
   cout << "Total = $" << total << endl;
   return 0;
}
```

Output

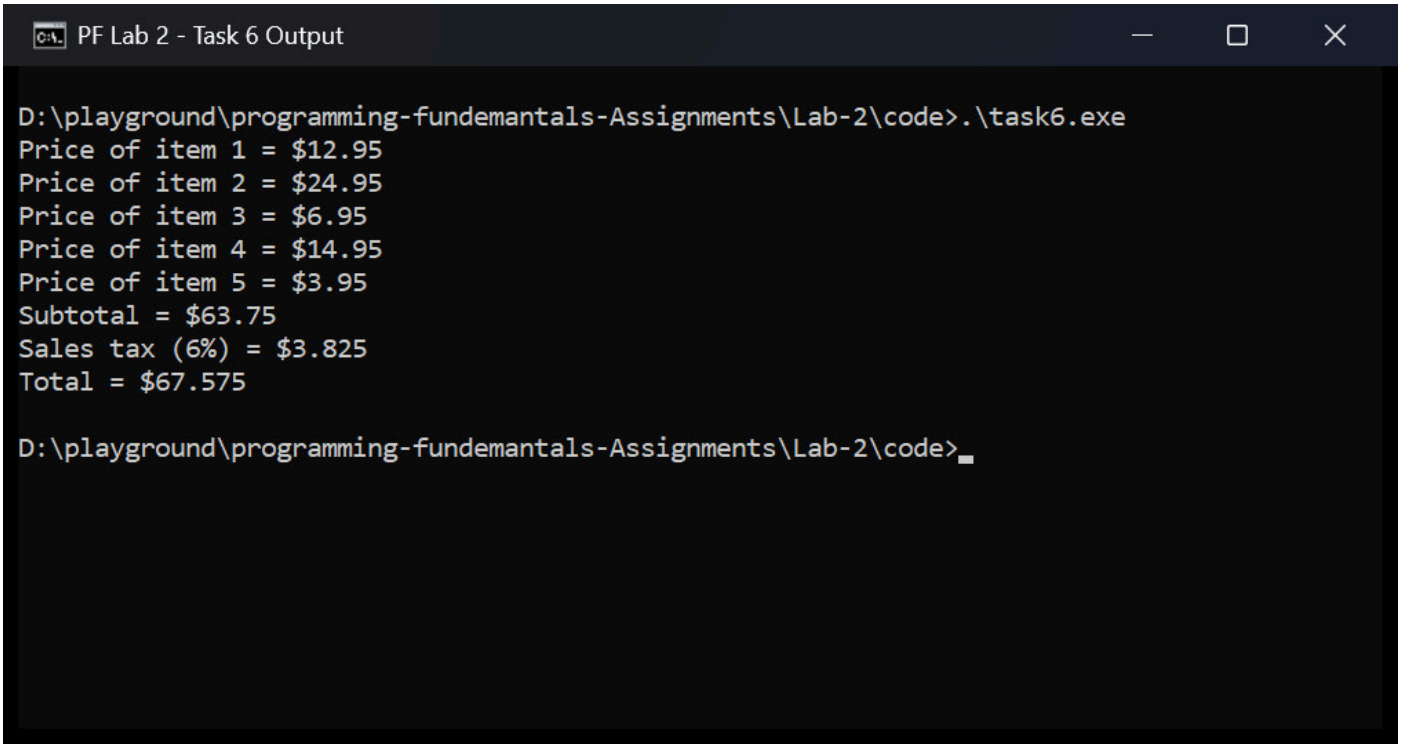
```
Price of item 1 = $ 12.95
Price of item 2 = $ 24.95
Price of item 3 = $ 6.95
Price of item 4 = $ 14.95
Price of item 5 = $ 3.95
Subtotal = $ 63.75
Sales tan (6%) = $ 3.825
Total = $ 67.575
```

Lab Task 6

Typed code (task6.cpp)

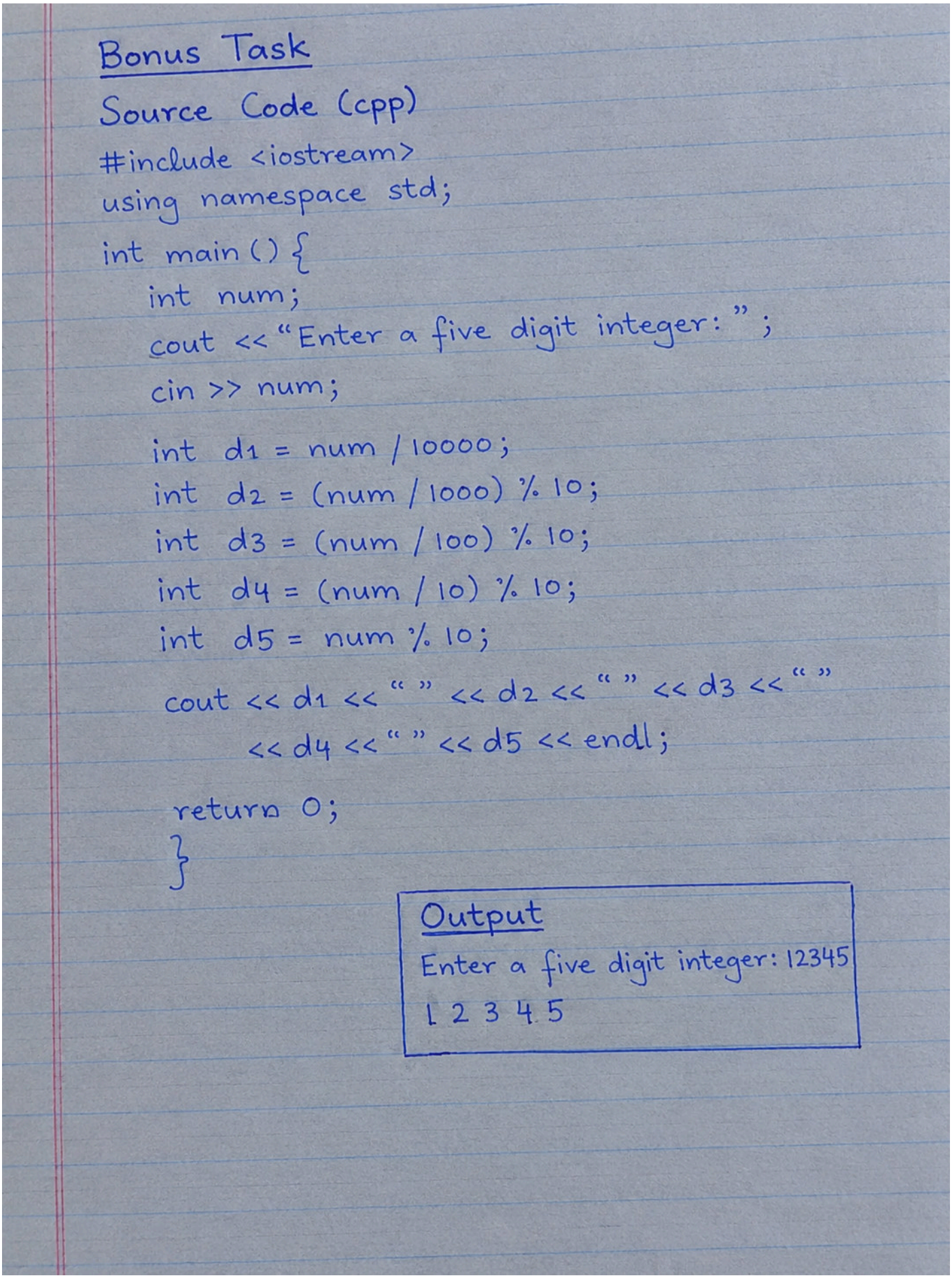


Output



Bonus Task

Handwritten code



Bonus Task

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main () {
    int num;
    cout << "Enter a five digit integer: " ;
    cin >> num;

    int d1 = num / 10000;
    int d2 = (num / 1000) % 10;
    int d3 = (num / 100) % 10;
    int d4 = (num / 10) % 10;
    int d5 = num % 10;

    cout << d1 << " " << d2 << " " << d3 << " "
         << d4 << " " << d5 << endl;

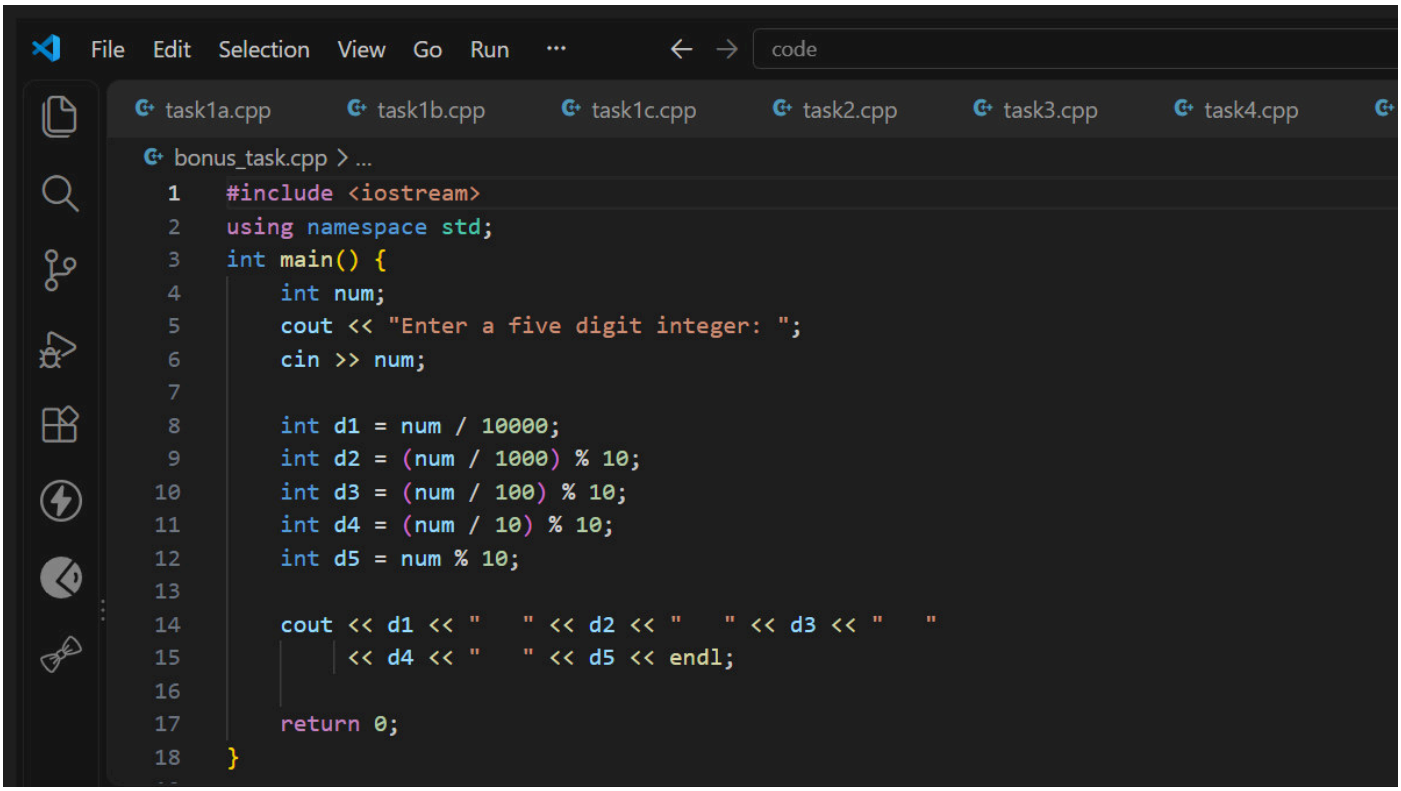
    return 0;
    }
```

Output

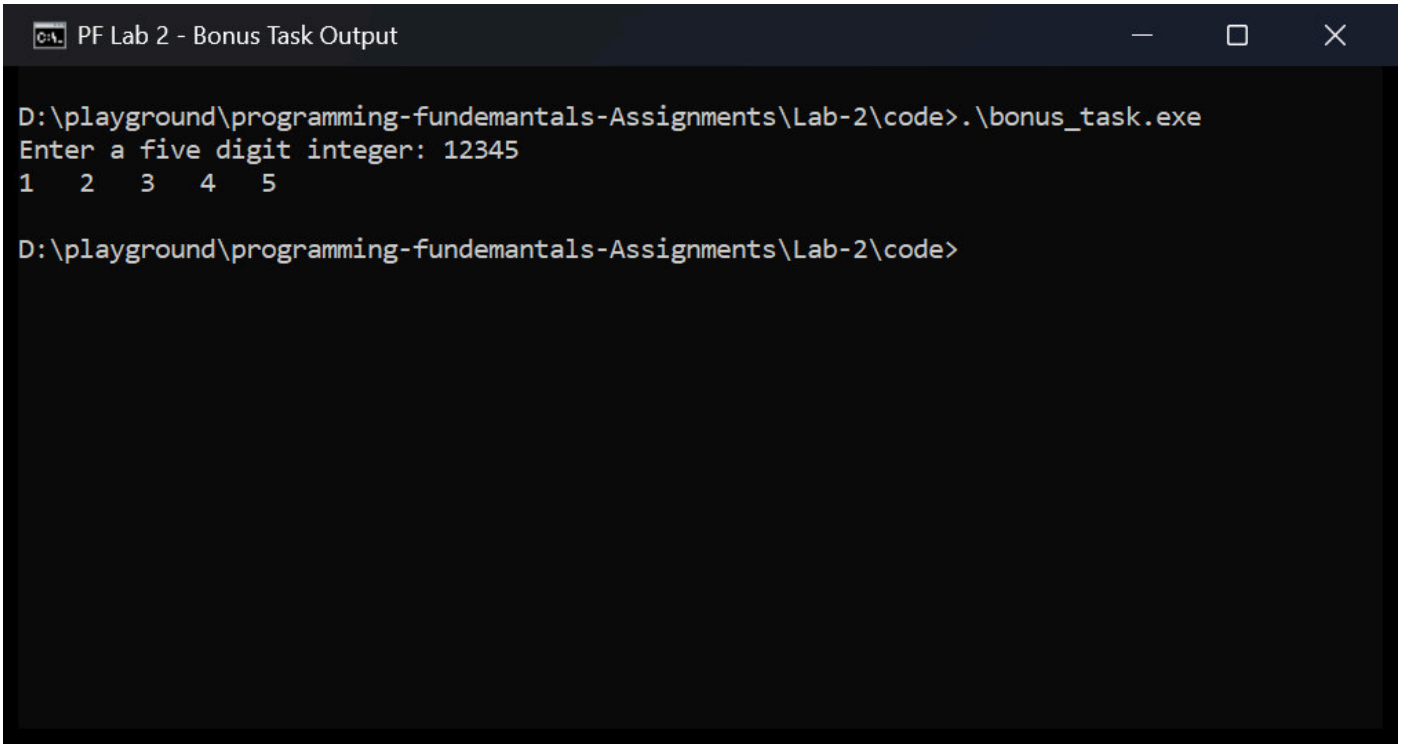
```
Enter a five digit integer: 12345
1 2 3 4 5
```

Bonus Task

Typed code (bonus_task.cpp)

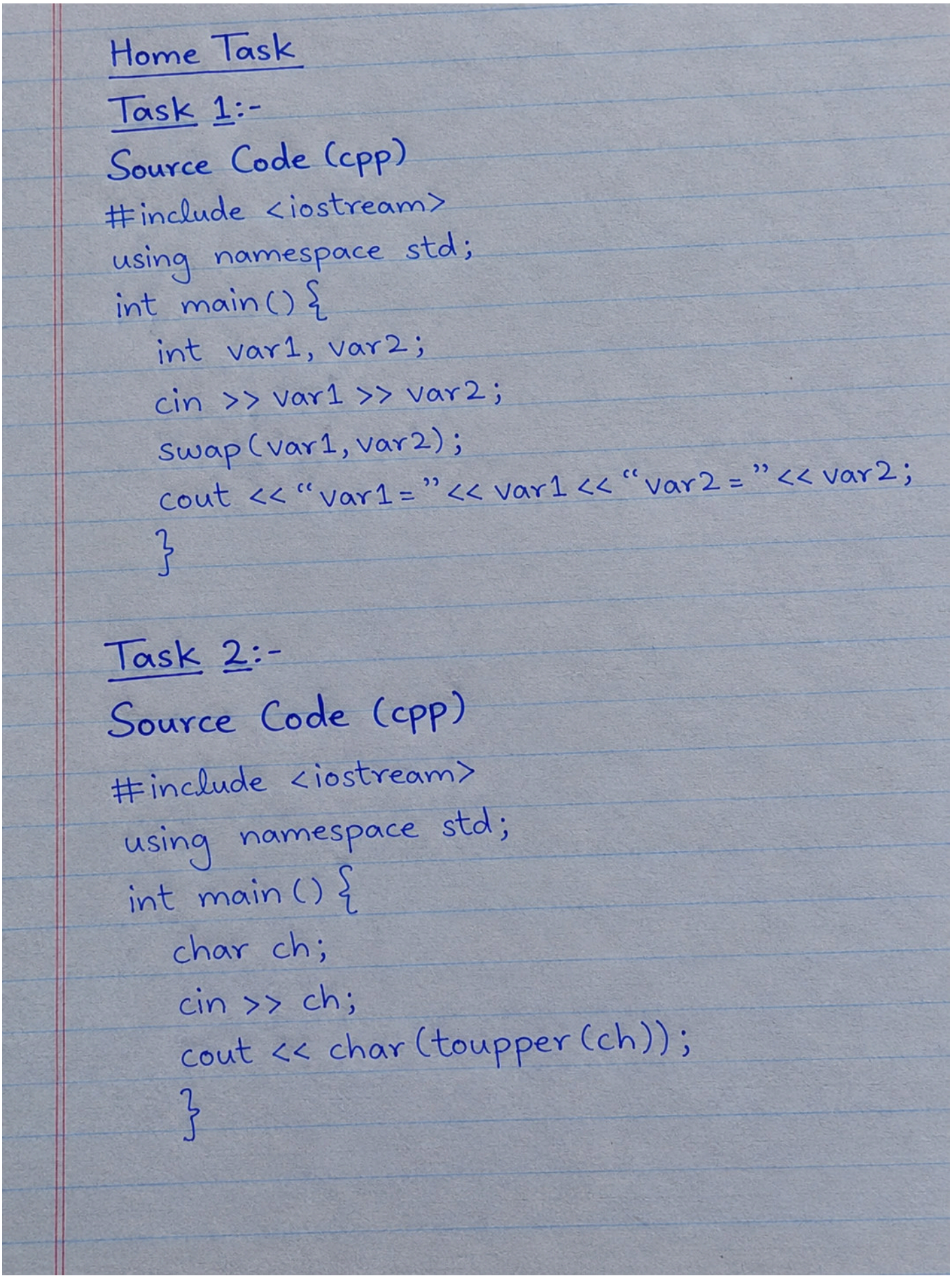


Output



Home Task 1

Handwritten code



Home Task

Task 1:-

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main() {
   int var1, var2;
   cin >> var1 >> var2;
   swap(var1, var2);
   cout << "var1 = " << var1 << "var2 = " << var2;
   }
```

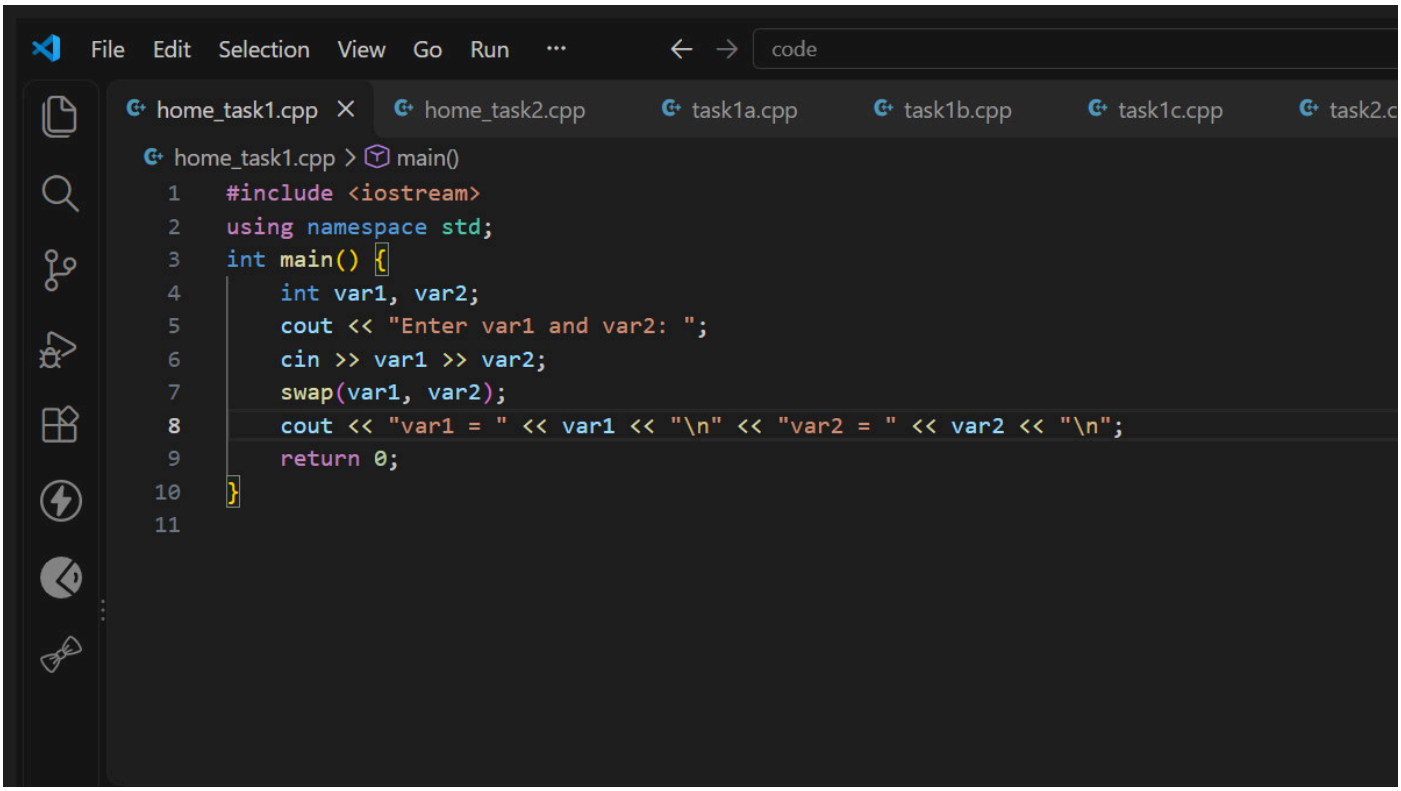
Task 2:-

Source Code (cpp)

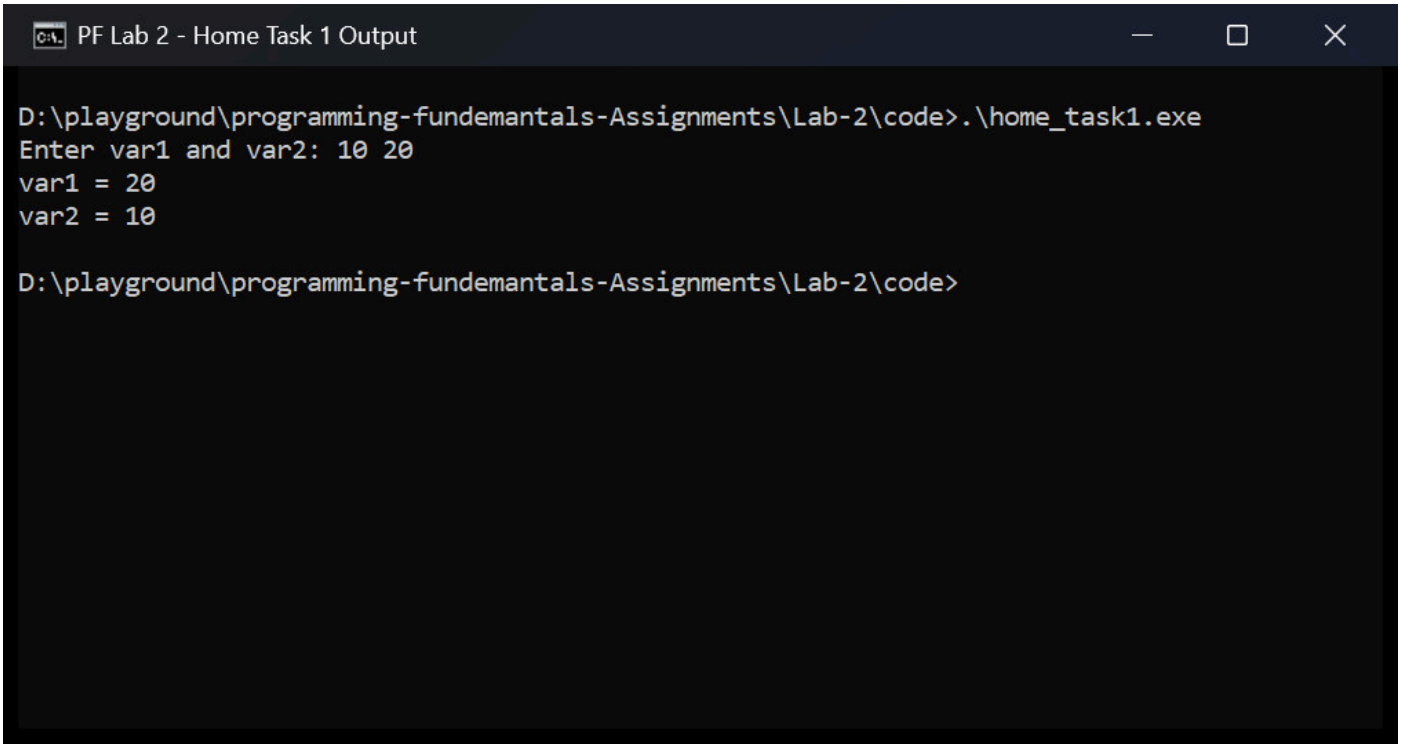
```
#include <iostream>
using namespace std;
int main() {
   char ch;
   cin >> ch;
   cout << char(toupper(ch));
   }
```

Home Task 1

Typed code (home_task1.cpp)

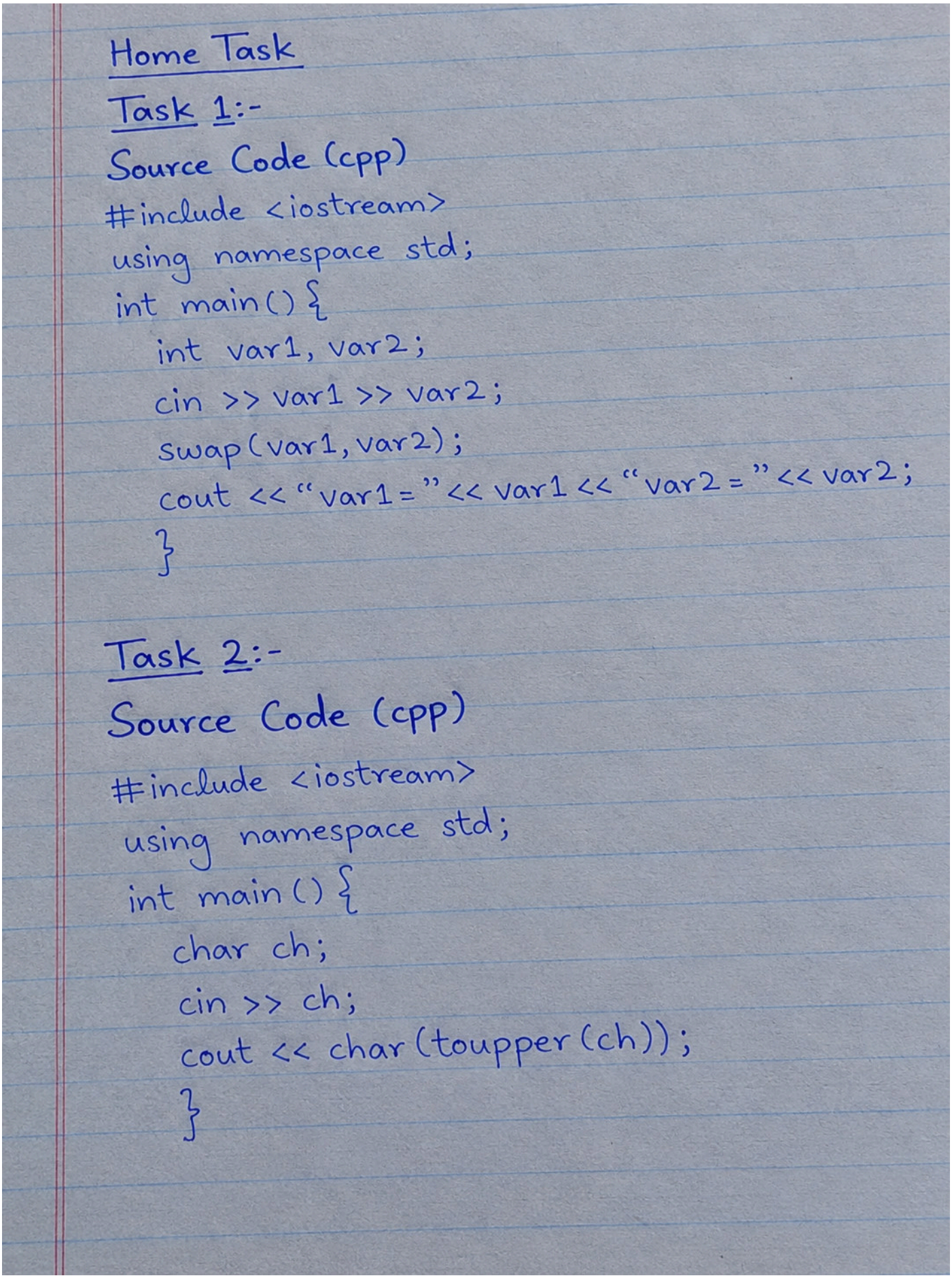


Output



Home Task 2

Handwritten code



Home Task

Task 1:-

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main() {
   int var1, var2;
   cin >> var1 >> var2;
   swap(var1, var2);
   cout << "var1 = " << var1 << "var2 = " << var2;
   }
```

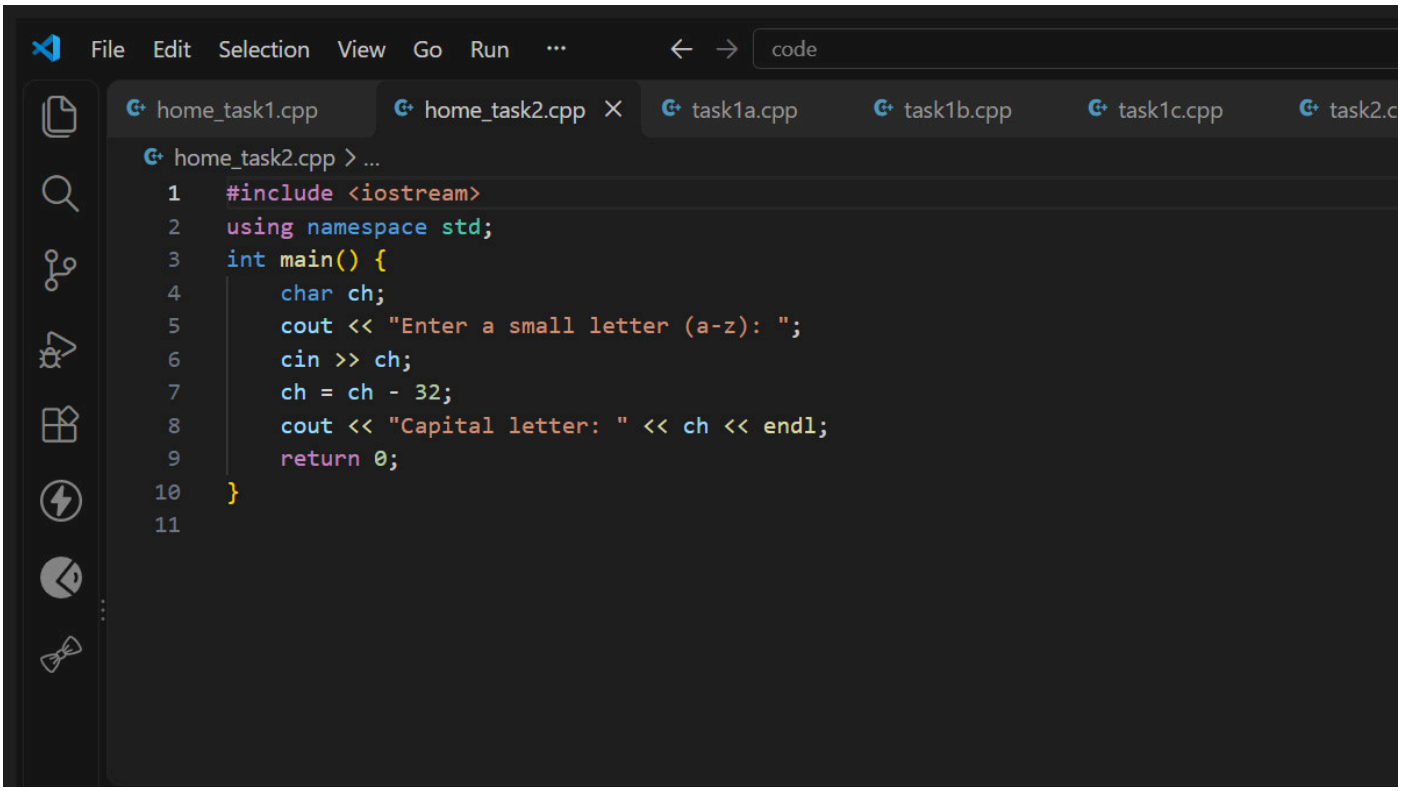
Task 2:-

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main() {
   char ch;
   cin >> ch;
   cout << char(toupper(ch));
   }
```

Home Task 2

Typed code (home_task2.cpp)



Output

